
OenoBench: A Wine-Domain Benchmark for Knowledge-Grounded Evaluation of Large Language Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

We introduce **OenoBench**, a benchmark of $\langle \approx 5,000 \rangle$ multiple-choice and short-answer questions designed to evaluate factual knowledge of large language models (LLMs) across the wine domain — a discipline that draws together botany, geography, chemistry, regulation, and sensory analysis, and which is uniquely well-suited to benchmark construction because it is supported by formal professional credentials (WSET, Master Sommelier, Master of Wine) whose syllabi externally validate what counts as expert knowledge. OenoBench is built from **38,104 atomic facts** extracted by 35 provenance-verified scrapers across 15 source types, organised into six domain pillars: wine regions, grape varieties, viticulture, winemaking, producers, and wine business. Questions are produced by five complementary generation strategies (fact-to-question, template, comparative, scenario, distractor mining) and distributed across five generator models (Claude, GPT, Gemini, Llama, deterministic templates) to mitigate single-model bias. Every question passes through a nine-agent automated audit organised into four teams (lexical hygiene, tri-judge correctness, deterministic checks, population statistics) and is calibrated against a human-reviewed gold sheet using Cohen’s κ . We report bias diagnostics including a *Self-Preference Score* that quantifies whether models score disproportionately well on questions generated by themselves, and we hold out a 300-question expert-authored control set for ground-truth comparison.

[PLACEHOLDER] Headline evaluation results across frontier LLMs: overall accuracy, per-domain accuracy, SPS values, control-set comparison.

1 Introduction

Large language models are increasingly evaluated on knowledge-intensive benchmarks, but the most widely-used resources — MMLU [1], GPQA [2], and their successors — share three properties that limit their usefulness as the field matures: (i) they aggregate breadth at the expense of *depth* within any single discipline, (ii) they are typically authored or curated by a small team using a single generation pipeline, exposing them to systematic blind-spots and stylistic regularities that frontier models can learn to exploit, and (iii) the questions are increasingly suspected of contamination through web-scale pre-training corpora. Recent work on benchmark contamination [6] and on self-preference bias in LLM-as-judge pipelines [16] suggests that as models grow more capable, the marginal value of a benchmark depends increasingly on *how* it was constructed, not only on what it contains.

We argue that domain-specialised benchmarks built around bodies of knowledge that are *externally validated by humans* — through certification, regulation, or peer-reviewed practice — offer a useful complement to broad knowledge benchmarks. Examples already exist in medicine [13], law [12],

34 and finance [9]. We extend this line of work to a domain that has so far received little attention from
35 the benchmarking community, despite being unusually well-defined: **wine**.

36 **Why wine.** Wine is a multi-disciplinary subject that integrates plant biology, geology and climate
37 science, organic chemistry, food microbiology, sensory neuroscience, regulatory law, and business
38 economics. It is one of very few domains for which (a) authoritative governmental and inter-
39 governmental sources publish structured taxonomic data (e.g. INAO, OIV, TTB, regional consortia),
40 (b) university research groups (notably UC Davis, Geisenheim, Bordeaux Sciences Agro) produce
41 open scholarly literature, and (c) professional certification bodies define a graded competence ladder
42 (WSET Levels 1–4, Court of Master Sommeliers, Institute of Masters of Wine) whose published
43 syllabi can be used to ground what “expert” should mean. Together these properties make it possible
44 to build a benchmark in which every question is traceable to a verifiable source, while still spanning
45 the recall–reasoning–application spectrum that distinguishes a strong taster, winemaker, or buyer
46 from a memorised glossary.

47 **Contributions.** This paper makes four contributions.

- 48 1. **A wine-domain benchmark** of $\langle \approx 5,000 \rangle$ questions covering six pillars (regions, vari-
49 eties, viticulture, winemaking, producers, business), built from 38,104 atomic facts whose
50 provenance is recorded down to the source URL and tier of authority.
- 51 2. **A multi-strategy, multi-model question-generation pipeline** (Section 3) that combines five
52 generation strategies and five generator models, with an explicit fact-grounding constraint:
53 the LLM’s role is reformatting verified facts, not providing them.
- 54 3. **A multi-agent quality audit** (Section 4) consisting of nine deterministic, statistical, and
55 tri-judge agents organised into four teams, calibrated against a human gold sheet using
56 Cohen’s κ and rejecting any agent signal whose human agreement falls below threshold.
- 57 4. **A bias-aware evaluation protocol** (Section 5) including held-out per-generator splits, an
58 expert-authored 300-question control set, and a *Self-Preference Score* that quantifies the
59 gap between a model’s accuracy on its own questions and on questions written by other
60 generators.

61 OenoBench, the source code for the construction and audit pipelines, the multi-agent audit reports,
62 and the human gold sheet are released under $\langle \text{license} + \text{URL pending de-anonymisation} \rangle$.

63 2 Related Work

64 **Broad knowledge benchmarks.** MMLU [1], BIG-Bench [5], AGIEval [3] and more recently
65 GPQA [2] and HLE [11] measure factual breadth across many subjects. As headline scores have
66 saturated for frontier models, attention has shifted toward harder reasoning benchmarks and toward
67 auditing the construction process [4]. Our work is complementary: rather than maximising difficulty,
68 we maximise depth within a single, externally-validated knowledge domain.

69 **Domain benchmarks.** MedQA [13], PubMedQA [15], LegalBench [12], and FinanceBench [9]
70 demonstrate the value of grounding evaluation in a regulated profession. We adopt their structure —
71 expert syllabus as taxonomy, source-traceable facts as backbone — and add multi-model question
72 generation and self-preference auditing as additional bias controls.

73 **LLM-as-judge bias and self-preference.** A growing literature documents that models tend to
74 favour outputs stylistically similar to their own [16, 14]. This biases benchmarks where the same
75 model generates and grades. Our pipeline distributes generation across five models and reports
76 per-model self-vs-other accuracy gaps as a first-class diagnostic.

77 **Benchmark contamination and solvability.** TODO [7, 8] show that questions present in pre-
78 training data inflate apparent capability. We mitigate this in two ways: (i) generating questions from
79 atomic facts rather than copying source text, and (ii) running an explicit *closed-book solvability*
80 check that flags questions answerable from world knowledge alone, allowing us to reserve them for a
81 separate evaluation slice.

Wine and beverage NLP. Prior NLP work on wine has focused on review-text generation [19], recommendation [18], and sensory descriptor extraction [17]. To our knowledge, OenoBench is the first benchmark targeting structured factual knowledge of viticulture, winemaking, and wine regions evaluated against frontier LLMs.

3 Dataset Construction

OenoBench is constructed in two stages: (1) a fact-collection stage that builds a corpus of atomic, provenance-tagged statements from authoritative sources; and (2) a question-generation stage that converts those facts into benchmark questions using five complementary strategies and five generator models. Both stages run inside a Postgres + Elasticsearch + Neo4j stack with full per-fact and per-question metadata, so that every benchmark item can be traced back to the URL, source tier, and generator that produced it.

3.1 Domain taxonomy

We organise wine knowledge into six *domain pillars*, chosen to align with the WSET Level 3 and Diploma syllabi:

1. **Wine regions** — appellations, classifications, geography, climate, soils.
2. **Grape varieties** — ampelography, parentage, regional distribution, sensory profile.
3. **Viticulture** — vineyard practice, training systems, pests/diseases, vintage effects.
4. **Winemaking** — vinification choices, fermentation, ageing, faults.
5. **Producers** — estates, négociants, co-operatives, key houses.
6. **Wine business** — markets, regulation, distribution, economics.

3.2 Source tiering and provenance integrity

Every fact is tagged with one of three source tiers, mirroring evidence-based-medicine practice:

- **Tier 1 (official, 19.6%):** government registries and inter-governmental bodies (INAO, OIV, TTB, Federdoc, UC Davis AVA database).
- **Tier 2 (authoritative, 76.6%):** Wikipedia (34.3%), Wikidata SPARQL (31.0%), wine consortia, peer-reviewed journals (*OENO One*, *Vitis*, *Australian Journal of Grape and Wine Research*).
- **Tier 3 (reliable, 3.8%):** curated open datasets on HuggingFace and Kaggle, secondary reference databases.

A 2026-04 internal audit identified 19 of an earlier 35-scraper set whose “scraped” facts were in fact hard-coded LLM-generated content. All 19 were rebuilt against live sources, and the entire 38,104-fact corpus reported here passes a per-source URL trace check. We document the rebuild because we believe **provenance auditing should be a first-class step in any LLM-built benchmark**: model-generated content masquerading as scraped data is a contamination vector that conventional dataset documentation does not detect.

Table 1: Atomic-fact corpus: 38,104 facts from 35 scrapers across 15 source types.

Domain pillar	Facts	Share
Wine regions	18,943	49.7%
Producers	6,215	16.3%
Grape varieties	5,959	15.6%
Viticulture	3,635	9.5%
Wine business	1,985	5.2%
Winemaking	1,367	3.6%
Total	38,104	100.0%

117 3.3 Atomic fact extraction

118 Facts are required to be (a) atomic — one assertion per sentence; (b) entity-tagged — with subject and
119 object linked to canonical IDs in our wine knowledge graph; and (c) source-faithful — a paraphrase of
120 the original, never a verbatim copy. A scraper-side validation pass enforces minimum length, predicate
121 presence, and near-duplicate suppression. Across ≈ 580 unique source URLs, post-deduplication
122 retention was `<retention %>`.

123 3.4 Question generation: five strategies, five models

124 A single-strategy, single-model benchmark inherits the blind-spots of its pipeline. We mitigate this
125 by combining five strategies and five generator models. Strategies and target shares:

- 126 • **Fact-to-question (40%)** — the LLM rewrites a single verified fact as a question; preserves
127 grounding but tends toward recall.
- 128 • **Template-based (25%)** — 45 deterministic, parameterised templates across the six pillars;
129 pure entity substitution, zero LLM creativity.
- 130 • **Comparative (15%)** — entity-affinity scoring pairs related entities (e.g. two Burgundy
131 premiers crus) and asks “which differs” or “what do both share”.
- 132 • **Scenario synthesis (10%)** — coherent fact clusters become applied multi-fact reasoning
133 prompts (e.g. a buyer scenario, a vintage decision).
- 134 • **Distractor mining (10%)** — confusable-entity sampling produces wrong answers that
135 share the right answer’s category and dimension, raising plausibility.

136 Generator distribution across the corpus is approximately Claude 30%, GPT 30%, Gemini 20%,
137 Llama 10%, deterministic templates 10%. A separate **300-question expert-authored control set**
138 (zero LLM involvement) is held back for use in evaluation (Section 5).

139 3.5 Pilot iteration

140 Generation has been iterated through pilot cycles `audit_pilot_v6` through `v13`, each comprising on
141 the order of 100–150 questions and a full audit pass. The current pilot, `audit_pilot_v13`, contains
142 146 questions (112 active, 34 in a `cb_reserve` pool flagged by the closed-book solvability gate) and
143 is the corpus on which Section 4 reports.

144 **[PLACEHOLDER]** Final 5,000-question generation run statistics: count, per-strategy share,
per-model share, retention rate, mean facts-per-question.

145 4 Multi-Agent Quality Audit

146 The audit pipeline runs every generated question through nine automated agents organised into four
147 teams. Each agent emits a per-question signal in `{PASS, WARN, FAIL}`, and signals are calibrated
148 against a human-reviewed gold sheet using Cohen’s κ . Agents whose human agreement falls below a
149 threshold are downweighted in aggregation, on the principle that an automated judge whose verdict
150 humans cannot reproduce should not gate question admission.

151 4.1 Team architecture

152 Five additional agents (C1 DistractorDifficulty, B3 ParaphraseStability, B4 Ambiguity, C3
153 SourceSwap, D2 DedupCalibration) are implemented but *escalation-gated*: they are invoked only
154 when an upstream agent flags a corpus-level concern. This deferred-execution policy keeps the
155 routine audit cost bounded while preserving full coverage when a defect class is suspected.

156 4.2 Gold-sheet calibration

157 A human gold sheet of 130 questions drawn from `audit_pilot_v13` is rated against eight rubrics:
158 `answer_correct`, `needs_source`, `no_vague_language`, `verbatim_copy`, `source_faithful`,
159 `wine_category_leak`, `distractors_plausible`, and an overall pass/fail. The reviewer is the
160 lead author (Diploma, WSET).

Table 2: Nine audit agents across four teams. Static agents use no LLM; tri-judge agents use a panel of three frontier models with majority voting.

Team	Agent	Function
A. Static	A1 LexicalHygiene	Regex sweep for vague phrasing, marketing language, meta-questions.
	A2 BiasStats	χ^2 position uniformity; Mann–Whitney U on answer length.
	A3 FactEcho	Longest-common-substring ratio between question text and source fact.
	A4 TemplateFingerprint	POS-bigram logistic regression detector (held-out AUC 0.84).
B. Tri-judge	B1 TriJudgeAnswer	Claude / GPT / Gemini consensus on the marked-correct answer.
	B2 ClosedBookSolvability	5-judge panel: can the question be answered from world knowledge alone?
C. Deterministic	C2 CategoryLeak	Wine-type distractor validation; flags impossible-category options.
D. Population	D1 SelfPreference	Per-model self-vs-other accuracy gap on the audited slice.
	D3 SkewAudit	Country / region over-representation ratio.

For each automated agent we compute Cohen’s κ against the corresponding human rubric. **Agent signals with $\kappa < 0.6$ are downweighted to advisory-only and excluded from gating decisions;** only signals that demonstrably reproduce expert judgement are allowed to gate question admission. This calibration step is run before every full generation cycle, so that drift in agent behaviour, rubric definitions, or upstream generator outputs is caught at the agent level rather than propagating into the released corpus.

[PLACEHOLDER] Per-agent κ table for the final corpus.

4.3 Pilot results: audit_pilot_v13

Across 146 questions, the nine-agent ensemble produced 561 PASS / 125 WARN / 66 FAIL signals. Headline findings:

- **B2 (ClosedBookSolvability):** 45 fails (30.8% of items). The most impactful defect class: nearly a third of the pilot is solvable from world knowledge alone without consulting the supporting fact. These items are not discarded; they are moved to a `cb_reserve` pool and used in a separate evaluation slice (Section 5) where pure recall is the explicit target.
- **A4 (TemplateFingerprint):** 9 of 19 templates flagged (47.4%). The held-out POS-bigram classifier reaches AUC 0.84 on distinguishing template-generated from human-written questions. Templates exceeding the threshold are sent to a paraphrase pass.
- **A3 (FactEcho):** 6 fails (4.1%) and 59 warns. Verbatim source copying is rare but not absent; flagged items are returned to generation with a stronger paraphrase prompt.
- **D3 (SkewAudit):** maximum country over-representation ratio of $5.84\times$ relative to a uniform baseline at $n = 146$. We expect this to attenuate sharply when the corpus reaches the 5,000-question target, but we report it explicitly because uneven domain coverage at small N otherwise inflates per-region accuracy estimates.

[PLACEHOLDER] Table: per-agent pass / warn / fail counts and κ vs. gold sheet for the full 5,000-question corpus.

4.4 Go / no-go gating

A pilot is promoted to a full generation run only when (a) the closed-book solvability fail rate is below 50%, (b) the template-fingerprint AUC is below 0.90 on held-out paraphrased items, (c) the per-country skew ratio is below $3\times$, and (d) human-rated overall pass rate on the gold sheet is above 80%.

[PLACEHOLDER] Final v13 go/no-go decision and resulting full-corpus generation outcomes.

5 Evaluation Protocol and Results

The evaluation protocol is designed to detect three failure modes simultaneously: (a) raw factual ignorance, (b) self-preference bias from generator overlap, and (c) inflation due to questions solvable from world knowledge alone.

5.1 Splits

We define four evaluation slices over the OenoBench corpus:

- **Full corpus** ($\langle N \approx 5,000 \rangle$) — the headline benchmark.
- **Per-generator held-out splits** — for each generator model g , the slice of questions *not* produced by g . Used to compute self-vs-other gaps.
- **Closed-book reserve** (`cb_reserve`) — questions flagged by agent B2 as world-knowledge solvable. Reported separately as a pure-recall slice.
- **Expert control set** ($N = 300$) — the WSET-Diploma-authored, zero-LLM control. Functions as a ground-truth comparator against which we calibrate the rest of the corpus.

5.2 Self-Preference Score

For each model m , we define

$$\text{SPS}(m) = \text{Acc}(m \mid \text{questions generated by } m) - \text{Acc}(m \mid \text{questions generated by other models}).$$

A pipeline that successfully neutralises generator bias should produce $|\text{SPS}(m)| < 3$ pp for every m in our model set. Larger gaps imply either (i) the generator’s own questions are systematically easier for it than for peers, or (ii) the audit pipeline is failing to remove stylistic fingerprints. We treat $|\text{SPS}| > 3$ pp as a corpus-level fail and trigger a re-paraphrase pass on the affected generator’s slice.

5.3 Models evaluated

We evaluate (*final list of frontier LLMs*) including: Claude (Opus 4.6, Sonnet 4.6), GPT (GPT-4o, GPT-5), Gemini (1.5 Pro, 3.1 Pro), Llama (3.1 405B), and at least one open small model as a baseline. Each model is run zero-shot, then with chain-of-thought prompting; we report the better of the two as the headline number, and both separately in the appendix.

5.4 Headline results

[PLACEHOLDER] Table 3: full-corpus accuracy per model, broken down by domain pillar (regions / varieties / viticulture / winemaking / producers / business). 8 rows, 7 columns. Include 95% bootstrap CI.

[PLACEHOLDER] Table 4: Self-Preference Score per model on the per-generator held-out slice. Pass threshold $|\text{SPS}| < 3$ pp.

[PLACEHOLDER] Table 5: closed-book reserve vs. active-corpus accuracy gap per model. The closed-book reserve is, by construction, easier; we expect a +X pp gap, and use deviations from this expectation to flag overfitting to the rest of the corpus.

[PLACEHOLDER] Figure 1: per-domain accuracy radar chart, all frontier models overlaid.

5.5 Comparison to expert control set

[PLACEHOLDER] Table 6: model accuracy on the 300-question expert control set vs. the LLM-generated bulk of the corpus. A model that scores significantly differently on the two slices reveals that LLM-generated questions are either easier (generation-side blind-spots) or harder (artefacts) than expert-written ones; the gap is itself a benchmark-validity diagnostic.

5.6 Error analysis

[PLACEHOLDER] Qualitative error analysis: 200 randomly-sampled wrong answers, hand-categorised by error type (factual confusion, regional misattribution, vintage error, regulatory misreading, sensory descriptor mismatch). Discussion of which domains and which generation strategies surface which error classes most often.

6 Limitations

Language and cultural scope. OenoBench is currently English-only and inherits the bias of its primary sources toward European (especially French, Italian, Spanish) and New-World canonical regions. Emerging wine cultures (e.g. China, India, much of Eastern Europe) are under-represented. The Tier-2 source mix relies heavily on Wikipedia and Wikidata, both of which are themselves English-skewed. Multilingual extension is the largest single gap.

Snapshot vs. a moving target. Wine regulation, classifications, and producer ownership change on a multi-year cadence (e.g. INAO appellation revisions, DOCG promotions, estate sales). The benchmark is a snapshot dated **<snapshot date>** and will require periodic re-extraction. We release the scrapers and provenance metadata so that re-extraction is mechanically reproducible.

Fact-grounding does not eliminate memorisation risk. Even though every question is generated from a verified atomic fact rather than a copied source paragraph, frontier LLMs may have seen the underlying facts in pre-training. The closed-book solvability slice (Section 4) is our principal mitigation: it isolates and reports separately the items that are answerable from world knowledge alone. We do not claim the active corpus is fully un-memorisable.

Single human reviewer. The current gold sheet is rated by the lead author. We plan to expand to a panel of three WSET-Diploma-or-higher reviewers before the full release; inter-rater κ will then be reported as a first-class quality metric alongside agent-vs-human κ .

Distribution shift between pilot and full release. Results in Section 4 are reported on a 146-question pilot. Behavioural differences are expected at the 5,000-question scale, particularly in skew metrics.

[PLACEHOLDER] Comparison of pilot vs. full-corpus headline metrics once the full run completes.

7 Ethics and Responsible Use

Alcohol context. Wine is an alcoholic beverage. The benchmark is a knowledge evaluation tool, not a consumer-facing product. Models that score highly on OenoBench should not be deployed as drinking-recommendation systems without separate alignment work covering health information, age-gating, and jurisdiction-specific advertising rules.

Source licensing. Tier-1 government sources are public records. Tier-2 Wikipedia and Wikidata content is used under CC-BY-SA / CC0; OenoBench question texts are derivative works produced by paraphrasing facts, not copies of source paragraphs, and are released under **<license TBD>**. Per-fact source URLs are released alongside the corpus so downstream users can verify and re-attribute.

Personal data. The corpus contains entity names of producers, winemakers, and historical figures who are public-record principals of wine estates. No private individuals are identified. The expert-authored control set contains no personal information; the reviewer is identified only as the lead author.

Misuse considerations. A wine-knowledge benchmark is low risk for direct misuse. The closest plausible misuse is using high-OenoBench-score models to automate fraudulent product descriptions or counterfeit provenance claims. We note this risk for completeness but do not see a benchmark-side mitigation distinct from general LLM-deployment safeguards.

References

- [1] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021.
- [2] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level google-proof q&a benchmark. In *Conference on Language Modeling (COLM)*, 2024.
- [3] TODO. Todo: Agieval citation, TODO.
- [4] TODO. Todo: benchmark construction audit, TODO.
- [5] TODO. Todo: Big-bench citation, TODO.
- [6] TODO. Todo: benchmark contamination overview, TODO.
- [7] TODO. Todo: contamination paper 1, TODO.
- [8] TODO. Todo: contamination paper 2, TODO.
- [9] TODO. Todo: Financebench citation, TODO.
- [10] TODO. Todo: Datasheets for datasets (gebru), TODO.
- [11] TODO. Todo: Humanity’s last exam citation, TODO.
- [12] TODO. Todo: Legalbench citation, TODO.
- [13] TODO. Todo: Medqa citation, TODO.
- [14] TODO. Todo: pandering / sycophancy in judges, TODO.
- [15] TODO. Todo: Pubmedqa citation, TODO.
- [16] TODO. Todo: Llm self-preference / judge bias, TODO.
- [17] TODO. Todo: sensory descriptor extraction, TODO.
- [18] TODO. Todo: wine recommendation systems, TODO.
- [19] TODO. Todo: wine review generation nlp, TODO.

A Datasheet for OenoBench

We follow the datasheet template of TODO [10].

A.1 Motivation

For what purpose was the dataset created? To evaluate factual knowledge of large language models on a multi-disciplinary, externally-validated domain (wine), and to provide a benchmark whose construction includes explicit bias controls (multi-model generation, multi-agent audit, self-preference scoring, expert-authored control set).

Who created the dataset?

[PLACEHOLDER] De-anonymised author and affiliation; the lead author holds the WSET Diploma in Wines.

Funding.

[PLACEHOLDER] Funding sources, hidden in anonymous submission.

296 A.2 Composition

297 **What do the instances represent?** Each instance is a benchmark question with: question text,
298 multiple-choice options (where applicable) with marked-correct answer, supporting fact(s) with
299 source URL and tier, generator model identifier, generation strategy, audit-agent verdicts, and (for the
300 gold-sheet subset) human reviewer ratings.

301 **How many instances are there in total?**

[PLACEHOLDER] Final question count, target 5,000. Pilot at submission time: 14 in audit_pilot_v13 (112 active, 34 cb_reserve).

302 **What data does each instance consist of?** Structured records over Postgres with columns described
303 in Section 3.

304 **Are there recommended data splits?** Yes, four (Section 5): full corpus, per-generator held-out,
305 closed-book reserve, expert control.

306 **Errors, noise, redundancies?** Documented in Section 4. Per-agent fail rates and κ versus the
307 human gold sheet are released alongside the corpus.

308 **Self-contained, or relies on external resources?** Self-contained: question text, supporting facts,
309 and source URLs are all included. Source URLs may rot over time; we mitigate by archiving Tier-1
310 source pages and recording the access timestamp.

311 A.3 Collection process

312 **How was the data acquired?** 35 provenance-verified web scrapers (released as src/scrapers/).
313 Sources span government registries, inter-governmental bodies (OIV), university research groups,
314 Wikipedia/Wikidata, peer-reviewed journals, and curated open datasets. Scraping was rate-limited
315 and used a research-disclosed user agent.

316 **Time frame.** Fact extraction: 2026-03 through 2026-04. Question generation: 2026-04 through
317 2026-05.

318 **Ethical review.**

[PLACEHOLDER] IRB / ethics-board status, if any.

319 A.4 Preprocessing / cleaning / labelling

320 Atomic-fact extraction with entity tagging; per-source paraphrase to avoid verbatim copy; near-
321 duplicate suppression; nine-agent automated audit; human gold-sheet rating on a 130-question
322 subset.

323 A.5 Uses

324 **Recommended uses.** Evaluation of LLM factual knowledge in the wine domain; ablation of bias-
325 mitigation techniques in benchmark construction; study of self-preference effects in LLM-as-judge
326 pipelines.

327 **Discouraged uses.** Direct deployment as a wine-recommendation or purchasing system without
328 separate alignment and age-gating; fine-tuning on the benchmark itself for ranking purposes (would
329 invalidate the benchmark for the trained model).

330 A.6 Distribution

331 **Will the dataset be distributed?** Yes, under (license TBD). Hosting: (HuggingFace Datasets URL
332 pending de-anonymisation).

333 **Citation.**

[PLACEHOLDER] BibTeX entry after de-anonymisation.

334 **A.7 Maintenance**

335 **Maintainer.** [PLACEHOLDER] Maintainer name and contact.

336 **Update cadence.** Wine regulation changes on a multi-year cadence; we plan annual re-extraction.
337 Scraper code is released so the community can re-extract independently.

338 **Errata.** A public errata log will be maintained at (URL TBD); corrections to questions, facts, or
339 audit verdicts will be versioned.

340 **B Additional Methodology Detail**

341 **B.1 Audit agent implementations**

342 [PLACEHOLDER] Per-agent algorithmic detail, prompt templates for tri-judge agents, threshold values for each warn/fail boundary, training data and held-out AUC for the A4 TemplateFingerprint classifier.

343 **B.2 Generation prompts**

344 [PLACEHOLDER] Per-strategy generation prompts (fact-to-question, comparative, scenario, distractor mining), with per-model temperature and top- p settings.

345 **B.3 Full per-domain evaluation tables**

346 [PLACEHOLDER] Per-pillar, per-model, per-prompting-strategy accuracy tables with bootstrap CIs.

347 **B.4 Per-country and per-region skew detail**

348 [PLACEHOLDER] Distribution of facts and questions by country and region; over-representation ratios; comparison to OIV global production statistics as an external baseline for “neutral” coverage.

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